Optimizing Real-Time Leaderboard Ranking Using Heap Structures

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Course: CS 5329 – Algorithm Design and Analysis

1. Introduction :

Modern digital platforms such as YouTube, Netflix, and gaming systems continuously rank items based on popularity and engagement. Efficient ranking of top items is essential for performance and user experience. This project compares two approaches for solving the Top-K ranking problem: a baseline sorting algorithm and an optimized heap-based algorithm.

2. Problem Statement:

The system processes a dataset of videos and ranks them based on an engagement score derived from views, likes, and comments. The goal is to efficiently retrieve the Top-K highest-ranked videos.

3. Algorithms Implemented :

a) Baseline Algorithm (Full Sorting)

The baseline approach sorts the entire dataset based on the engagement score and selects the top K items.

- Time Complexity: **O(n log n)**
- Space Complexity: **O(n)**

b) Optimized Algorithm (Min-Heap)

The optimized approach uses a min-heap of size K. It iterates through the dataset and maintains only the top K elements.

- Time Complexity: **O(n log k)**
- Space Complexity: **O(k)**

4. Dataset Description :

A synthetic dataset was generated to simulate video engagement data. The dataset contains:

- title
- views
- likes
- comment count

The dataset sizes used:

- 10,000 records
- 50,000 records
- 100,000 records

5. Runtime Results :

Dataset Size	Baseline Runtime (s)	Heap Runtime (s)
Sample	0.000157	0.000079
10k	0.001854	0.000680
50k	0.005784	0.002065
100k	0.012792	0.003801

6. Memory Usage:

Dataset Size	Baseline Memory (MB)	Heap Memory (MB)
Sample	0.0230	0.0002
10k	0.2290	0.0003
50k	1.1444	0.0003
100k	2.2886	0.0003

7. Conclusion :

This project demonstrates that the heap-based approach is more efficient for Top-K ranking problems. It provides better scalability, lower memory usage, and faster execution compared to full sorting, especially for large datasets.